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基於潛在擴散模型之訊號分離與重建於多用戶語義衛星通訊

Latent Diffusion-Based Signal Separation and Reconstruction for Multi-User Semantic Satellite Communications

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摘　要

隨著未來無線通訊系統對高效率資訊傳輸需求的提升，語意通訊（Semantic Communication）逐漸成為重要的研究方向，其核心概念在於傳輸與任務相關的語意資訊，而非追求符號層級的完全重建。然而，在多使用者語意通訊系統中，不同使用者的語意表示經由共享無線通道傳輸時可能產生疊加，進而造成使用者間的語意干擾並降低重建品質。

為了解決此問題，本論文提出一種目標感知語意分離（Target-Aware Semantic Separation, TASS）架構，用於多使用者語意通訊系統中的目標特定潛在表示恢復。在傳送端，利用變分自編碼器（Variational Autoencoder, VAE）將輸入影像映射至低維度的語意潛在表示，並於傳輸前進行全域尺度調整。在接收端，採用條件式去噪擴散機率模型（Conditional Denoising Diffusion Probabilistic Model, DDPM）作為潛在表示分離器，從接收到的混合潛在表示中恢復指定使用者的語意表示。透過將接收端的混合潛在表示與目標條件共同作為反向擴散過程的條件資訊，使分離器能夠針對指定使用者進行目標特定的潛在表示恢復。完成分離後，恢復的潛在表示再透過具梯度懲罰之條件式 Wasserstein 生成對抗網路（Conditional WGAN-GP）重建至影像域。

為提升訓練穩定性，本研究採用兩階段訓練策略，首先建立穩定且具有良好影像重建能力的語意潛在空間，再固定語意編碼器與生成器的參數，進一步訓練條件式 DDPM 潛在表示分離器。本研究使用 EuroSAT 與 DeepGlobe 衛星影像資料集進行實驗，並在不同訊號雜訊比、干擾強度、干擾使用者數量以及無線通道模型下評估所提出的方法。實驗結果顯示，TASS 能有效降低多使用者語意干擾所造成的影響，並在中度至強干擾環境下展現較佳的影像重建穩健性。研究結果證明，透過目標感知的潛在域分離進行多使用者語意干擾抑制具有可行性，並可作為未來多使用者語意通訊系統設計的潛在方向。

關鍵字— 語意通訊、多使用者通訊、語意干擾、去噪擴散機率模型、潛在表示分離、變分自編碼器、生成對抗網路

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Abstract

Semantic communication has emerged as a promising paradigm for future wireless communication systems by focusing on the transmission of task-relevant semantic information rather than exact symbol-level recovery. However, in multi-user semantic communication systems, semantic representations transmitted over shared wireless channels may become superimposed, resulting in inter-user semantic interference and degraded reconstruction quality. To address this issue, this thesis proposes a Target-Aware Semantic Separation (TASS) framework for target-specific latent recovery in multi-user semantic communication systems. At the transmitter, a variational autoencoder (VAE)-based semantic encoder maps input images into compact latent representations, which are globally scaled before transmission. At the receiver, a conditional denoising diffusion probabilistic model (DDPM)-based latent separator is employed to recover the desired user-specific latent representation from the received latent mixture. By jointly conditioning the reverse diffusion process on the received mixture and a target condition, the proposed separator can selectively recover the semantic representation associated with the desired user. The recovered latent representation is subsequently reconstructed in the image domain using a conditional Wasserstein generative adversarial network with gradient penalty (WGAN-GP). A two-stage training strategy is adopted to first establish a stable semantic latent space and then train the conditional latent separator while keeping the semantic encoder and generator fixed. Experiments using the EuroSAT and DeepGlobe satellite image datasets evaluate the proposed framework under different signal-to-noise ratios, interference strengths, numbers of interferers, and wireless channel models. The results demonstrate that the proposed TASS framework effectively mitigates multi-user semantic interference and provides improved reconstruction robustness, particularly under moderate-to-strong interference conditions. These results demonstrate the feasibility of target-aware latent-domain separation as an approach to interference mitigation in multi-user semantic communication systems.

Keywords— Semantic communication, multi-user communication, semantic interference, denoising diffusion probabilistic model, latent separation, variational autoencoder, generative adversarial network

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Chapter 1

Introduction

Next-generation wireless communication systems are expected to support high data throughput and reliable transmission. With the development of the fifth-generation (5G) and the upcoming sixth-generation (6G) wireless communication systems, network traffic and communication requirements are expected to continue increasing [1, 2]. Many emerging applications, such as autonomous vehicles, extended reality, and the Internet of Things, require high data rates, low transmission latency, and high reliability [3, 1]. In addition, the increasing number of connected devices generates a large amount of data that needs to be transmitted through wireless networks. These increasing requirements create additional challenges for future wireless communication systems. Traditional communication systems mainly focus on accurately transmitting information from the transmitter to the receiver. To provide reliable communication, channel coding and other error protection methods are commonly used to reduce the effect of channel noise and transmission errors. However, transmitting and protecting a large amount of data requires additional communication resources. At the same time, spectrum and energy resources in wireless communication systems are limited. As the number of users and the amount of transmitted data continue to increase, efficiently using the available communication resources becomes increasingly important. Therefore, efficiently transmitting useful information under limited communication resources has become an important topic for next-generation wireless communication.

Semantic communication has been proposed as a possible solution to meet these increasing communication requirements. Unlike traditional communication systems, which aim to accurately recover every transmitted bit, semantic communication focuses on recovering the meaning of the transmitted information [4, 5, 6, 7]. In a semantic communication system, the transmitter extracts important semantic features from the source data and transmits

them through a wireless channel. The receiver then uses the received semantic features to reconstruct the source information or complete a specific task. Xie et al. proposed DeepSC, an end-to-end semantic communication system for text transmission [8]. DeepSC uses a Transformer-based encoder to extract semantic information from sentences and jointly optimizes semantic and channel coding. Their results showed that DeepSC can maintain better semantic similarity than traditional communication methods, especially under low signal-to-noise ratio conditions. However, DeepSC mainly considers point-to-point text transmission and does not address semantic interference between multiple users. For wireless image transmission, Bourtsoulatze et al. proposed DeepJSCC, which uses an autoencoder to directly map input images into channel symbols [9]. The encoder, wireless channel, and decoder are jointly optimized during model training, and DeepJSCC provides graceful performance degradation when the channel condition becomes worse. However, DeepJSCC mainly focuses on image transmission over a noisy channel and does not consider the separation of semantic features transmitted by different users. Erdemir et al. further proposed GenerativeJSCC and InverseJSCC by introducing generative models into semantic image transmission [10]. These methods use generative models and perceptual loss functions to improve the visual quality of reconstructed images under limited bandwidth and low signal-to-noise ratio conditions. However, improving perceptual quality may introduce a tradeoff between visual realism and the fidelity of the reconstructed image to the original input. Furthermore, these methods mainly consider point-to-point transmission rather than the recovery of individual semantic features in a multi-user environment.

When semantic communication is extended to a multi-user environment, multiple users may share the same wireless resources to transmit their semantic information. Recent studies have investigated multi-user semantic communication from different perspectives [11, 12]. Wei et al. proposed a task-oriented multi-user semantic communication framework based on deep over-the-air computation to support the aggregation of semantic information from multiple users [13]. Li et al. investigated semantics-empowered non-orthogonal multiple access for the downlink transmission of correlated information sources, demonstrating the potential of exploiting semantic correlation for efficient multi-user transmission [14]. In addition, multi-user semantic communication has been investigated for semantic information fusion over shared wireless channels [15].

Although these studies demonstrate the potential of semantic communication in multi-user environments, their primary objectives are semantic aggregation, information fusion, or efficient resource sharing rather than the explicit recovery of an individual user's semantic representation from a re-

ceived mixture. When multiple semantic representations are simultaneously transmitted over shared wireless resources, the received signal may contain not only channel noise but also structured semantic interference from other users. Therefore, recovering the desired semantic representation requires the receiver to distinguish the target semantic source from interfering semantic components. This motivates the development of semantic-domain interference management and user-specific semantic recovery techniques for multi-user communication systems.

Denoising diffusion probabilistic models (DDPMs) provide a promising approach for recovering corrupted representations because of their strong denoising and data recovery capabilities. A DDPM gradually adds Gaussian noise to the original data through a forward diffusion process and learns to recover the original data through a reverse denoising process [16, 17]. Based on this denoising capability, Wu et al. proposed a channel denoising diffusion model to mitigate wireless channel distortion in semantic communication systems [18]. Their method employs a diffusion-based denoising process to recover semantic features corrupted by channel noise and improves semantic reconstruction under poor channel conditions. However, the method primarily focuses on channel denoising rather than separating semantic representations originating from different users. Zeng et al. further proposed a diffusion model-based channel enhancer for multi-user semantic communication [19]. Their method mitigates channel distortion and multi-user interference to improve the quality of fused semantic features for a downstream segmentation task. Nevertheless, the received semantic features are enhanced for feature fusion rather than explicitly separated into the individual latent representation of each user. Therefore, user-specific latent separation from a received mixture containing both channel noise and semantic interference remains an important problem. Table 1.1 summarizes the main differences between the existing methods and the proposed method.

As shown in Table 1.1, existing semantic communication methods have been developed for image transmission, generative reconstruction, channel denoising, and multi-user communication. However, existing diffusion-based methods mainly focus on channel denoising, feature enhancement, or semantic fusion. Although some methods consider inter-user interference in multi-user environments, user-specific latent separation from a received mixture has received limited attention. For multi-user semantic communication, directly reconstructing the received mixture may preserve interference from other users, while conventional denoising methods mainly remove channel noise without identifying the desired semantic source. Therefore, a separation method that can use user information to identify and recover the desired semantic representation is required. Although diffusion models require an

Table 1.1: Comparison of existing semantic communication methods and the proposed method.

Literature	[9]	[10]	[13]	[14]	[18]	[19]	**Proposed**
Image Transmission	✓	✓	–	–	✓	–	✓
Generative Reconstruction	–	✓	–	–	–	–	✓
Diffusion Model	–	–	–	–	✓	✓	✓
Multi-User	–	–	✓	✓	–	✓	✓
Inter-User Interference	–	–	–	–	–	✓	✓
Target-Conditioned Separation	–	–	–	–	–	–	✓
User-Specific Latent Separation	–	–	–	–	–	–	✓

iterative reverse denoising process and introduce additional computational complexity, their data recovery ability provides a possible solution for separating the desired latent representation from semantic interference. To address this problem, this thesis develops a multi-user semantic communication framework with a conditional DDPM-based latent separator. The source images are first encoded into latent representations and transmitted through a shared wireless channel. At the receiver, the conditional DDPM takes the received latent mixture and the target user label as conditions and recovers the desired latent representation through the reverse denoising process. The recovered latent representation is then reconstructed into an image by the corresponding decoder. In this way, the proposed method considers both channel noise and semantic interference from other users and performs user-specific separation directly in the latent domain. The main contributions of this thesis are summarized as follows:

- A multi-user semantic communication framework is developed to investigate semantic feature transmission under both channel noise and inter-user interference. Unlike conventional point-to-point semantic communication, the received latent representation contains semantic information from multiple users.

- A Target-Aware Semantic Separation (TASS) framework is proposed to recover the desired user-specific latent representation from the received mixture. The received latent mixture and the target-user condition are incorporated into the conditional reverse diffusion process, enabling TASS to recover the desired user-specific latent representation from the multi-user latent mixture.

- The proposed method is evaluated using two satellite image datasets, EuroSAT and DeepGlobe, under different channel and interference con-

ditions. Reconstruction performance is quantified using PSNR to evaluate the ability of the proposed separator to recover user-specific semantic information under different channel and interference conditions.

Chapter 2

System Model and Architecture

2.1 System Model

We consider a multi-user semantic communication system consisting of K transmitter–receiver pairs, as illustrated in Fig. 2.1. The set of users is denoted by

$$\mathcal{U} = \{1, 2, \ldots, K\}, \tag{2.1}$$

where K denotes the total number of users.

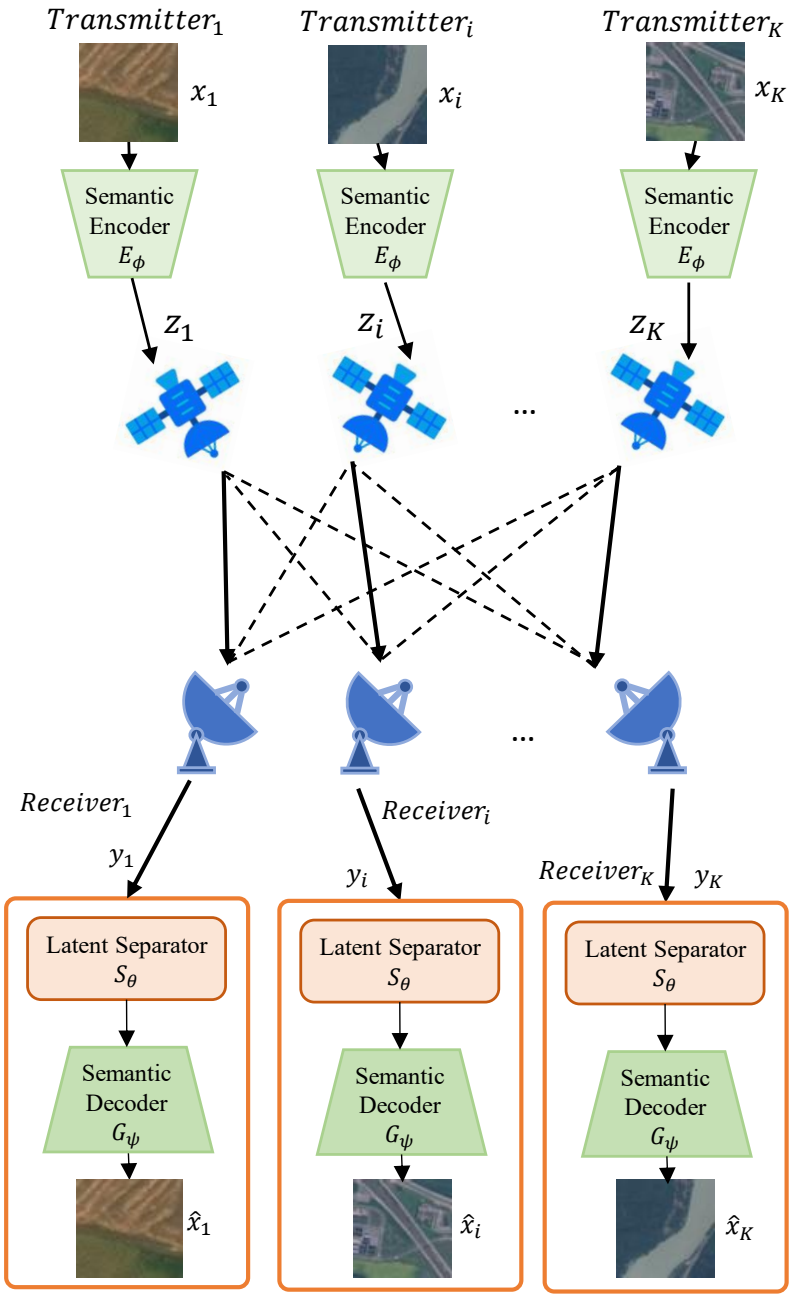


Figure 2.1: System model of the considered multi-user semantic communication system over satellite downlink channels. Solid lines represent the desired transmission links, while dashed lines represent interfering links from other users.

2.1.1 Multi-User Semantic Transmission Model

For each user $k \in \mathcal{U}$, let $\mathbf{x}_k$ denote the input semantic source. Although semantic information can be represented in different modalities, such as images, text, and audio, this work focuses on image transmission. Each input is mapped into a latent representation by a shared semantic encoder $E_\phi(\cdot)$ as

$$\mathbf{z}_k = E_\phi(\mathbf{x}_k), \tag{2.2}$$

where $\mathbf{z}_k \in \mathbb{R}^d$, d denotes the dimension of the latent representation, and ϕ denotes the shared encoder parameters. Before transmission, a fixed global scaling factor is applied to maintain a consistent latent scale across different semantic sources. The transmitted latent representation is expressed as

$$\tilde{\mathbf{z}}_k = \gamma \mathbf{z}_k, \tag{2.3}$$

where $\gamma > 0$ denotes a fixed global scaling factor shared by all users. Unlike sample-wise power normalization, the same scaling factor is applied to all latent representations, thereby preserving their relative latent-space structure.

In this work, we focus on a two-user scenario consisting of one target user and one interfering user, where each user is equipped with a single antenna. When both users simultaneously transmit their latent representations over a shared wireless channel, the received latent signal at the target receiver is modeled as

$$\mathbf{y}_k = h_k \tilde{\mathbf{z}}_k + \sum_{\substack{i=1 \\ i \neq k}}^{K} \alpha_i h_i \tilde{\mathbf{z}}_i + \mathbf{n}_k, \tag{2.4}$$

where h_k and h_i denote the effective channel gains associated with the target and interfering links, respectively, and $\alpha_i \geq 0$ denotes the interference amplitude coefficient.

2.1.2 Wireless Channel Model

The channel gains are modeled according to the considered propagation conditions. In particular, Rician fading is considered to represent the dominant line-of-sight propagation characteristics of a LEO satellite downlink [20]. The channel coefficient for user k is modeled as

$$h_k = \sqrt{\frac{K_R}{K_R + 1}}\, h_k^{\mathrm{LoS}} + \sqrt{\frac{1}{K_R + 1}}\, h_k^{\mathrm{NLoS}}, \tag{2.5}$$

where K_R denotes the Rician K-factor, while h_k^{LoS} and h_k^{NLoS} represent the line-of-sight and scattered components, respectively. The additive white

Gaussian noise (AWGN) is modeled as

$$\mathbf{n}_k \sim \mathcal{N}(\mathbf{0}, \sigma_k^2 \mathbf{I}), \tag{2.6}$$

where σ_k^2 denotes the noise variance.

The channel quality is characterized by the signal-to-noise ratio (SNR). Since the target latent representation is normalized to unit average power, the received SNR_k is defined as

$$\mathrm{SNR}_k = \frac{|h_k|^2}{\sigma_k^2}, \tag{2.7}$$

and its value in decibels is expressed as

$$\mathrm{SNR}_{k,\mathrm{dB}} = 10 \log_{10} \left(\frac{|h_k|^2}{\sigma_k^2} \right). \tag{2.8}$$

Similarly, the interference-to-signal power ratio (ISR) is defined as

$$\mathrm{ISR}_k = \frac{\sum\limits_{\substack{i=1 \\ i \neq k}}^{K} \alpha_i^2 |h_i|^2}{|h_k|^2}, \tag{2.9}$$

and the corresponding ISR in decibels is given by

$$\mathrm{ISR}_{k,\mathrm{dB}} = 10 \log_{10} \left(\frac{\sum\limits_{\substack{i=1 \\ i \neq k}}^{K} \alpha_i^2 |h_i|^2}{|h_k|^2} \right). \tag{2.10}$$

2.1.3 Target Semantic Recovery

The objective of the receiver is to recover the latent representation of the target user from the received mixture in (2.4). Unlike conventional multi-antenna signal separation, the received signal in (2.4) is a single latent mixture containing semantic representations from multiple users. Therefore, conventional zero-forcing separation based on an invertible channel matrix cannot be directly applied to recover the individual latent representations.

To address this problem, a user-conditioned latent separation function $S(\cdot)$ is introduced to estimate the desired latent representation as

$$\hat{\mathbf{z}}_k = S\left(\mathbf{y}_k, c_k\right), \tag{2.11}$$

where c_k denotes the target condition used to identify the desired semantic source. The estimated latent representation is subsequently mapped back to the semantic domain by a conditional semantic decoder $G(\cdot)$ as

$$\hat{\mathbf{x}}_k = G\left(\hat{\mathbf{z}}_k, c_k\right), \tag{2.12}$$

where $\hat{\mathbf{x}}_k$ denotes the reconstructed semantic source of the target user.

2.2 Problem Formulation

Based on the system model described in Section 2.1, the objective is to recover the target user's latent representation from the received mixture while suppressing the semantic components introduced by interfering users. Accordingly, the user-specific latent recovery problem is formulated as

$$\min_{\Theta} \quad \mathbb{E}\left[\left\|\tilde{\mathbf{z}}_k - \hat{\tilde{\mathbf{z}}}_k\right\|_2^2\right] \tag{2.13a}$$

$$\text{s.t.} \quad \hat{\tilde{\mathbf{z}}}_k = S\left(\mathbf{y}_k, c_k; \Theta\right), \tag{2.13b}$$

$$\mathbf{y}_k = h_k \tilde{\mathbf{z}}_k + \sum_{\substack{i=1 \\ i \neq k}}^{K} \alpha_i h_i \tilde{\mathbf{z}}_i + \mathbf{n}_k, \tag{2.13c}$$

$$\tilde{\mathbf{z}}_i = \gamma \mathbf{z}_i, \qquad i \in \mathcal{U}, \tag{2.13d}$$

$$\alpha_i \geq 0, \qquad i \neq k. \tag{2.13e}$$

The objective in (2.13) minimizes the distortion between the recovered latent representation and the target user's latent representation. The constraints describe the multi-user latent superposition, the transmit-power normalization, and the nonnegative interference strength. The specific realization of $S(\cdot)$ is introduced in Chapter 3.

Chapter 3

Proposed Methodology

3.1 Overview of the Proposed Framework

In a multi-user semantic communication system, semantic representations generated by different users may be simultaneously transmitted over a shared communication channel. As a result, the received latent representation contains not only the desired semantic information but also interference from other users and channel noise. Directly decoding such an interfered latent representation may lead to significant semantic distortion, since the decoder receives a representation that deviates from the latent distribution learned during training. To address this problem, we propose a Target-Aware Semantic Separation (TASS) framework for multi-user semantic communication. Instead of separating the transmitted signals in the pixel domain, TASS performs inter-user interference separation directly in the learned latent space. The overall architecture of the proposed TASS framework is illustrated in Fig. 3.1. For each user k, the input image $\mathbf{x}_k$ is first mapped into a compact semantic representation $\mathbf{z}_k$ by the pretrained VAE-based semantic encoder $E_\phi(\cdot)$. The latent representations are then globally scaled using a fixed scaling factor and transmitted through the shared channel, where the desired latent representation is superimposed with interference from other users and channel noise.

At the receiver, a conditional denoising diffusion probabilistic model (DDPM) is employed as the latent separator. Given the received latent mixture and the condition corresponding to the target user, the separator estimates the desired latent representation while suppressing the interfering semantic components. This conditional design enables a single separator to recover different semantic components from the same received mixture according to the specified target condition. The separation process can there-

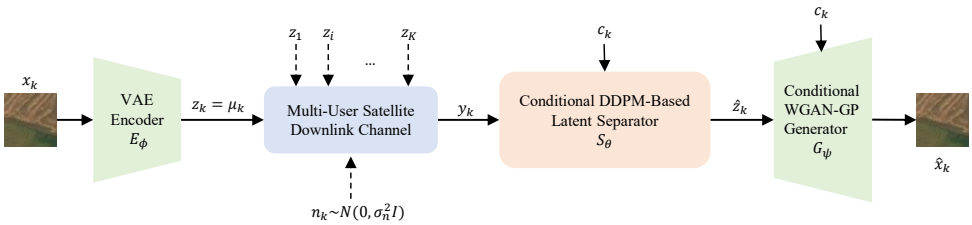


Figure 3.1: Overview of the proposed multi-user semantic communication framework over satellite downlink channels.

fore be expressed as

$$\hat{\tilde{\mathbf{z}}}_k = S_\theta\left(\mathbf{y}_k, c_k\right), \tag{3.1}$$

where $\mathbf{y}_k$ denotes the received latent mixture observed by target user k, c_k represents the target condition, and $S_\theta(\cdot)$ denotes the proposed conditional DDPM-based latent separator. After latent separation, the estimated scaled latent representation $\hat{\tilde{\mathbf{z}}}_k$ is restored to its original scale and passed to the conditional generative decoder to reconstruct the desired semantic source as

$$\hat{\mathbf{x}}_k = G_\psi\left(\frac{\hat{\tilde{\mathbf{z}}}_k}{\gamma}, c_k\right), \tag{3.2}$$

where $G_\psi(\cdot)$ denotes the conditional generative decoder. By performing separation before semantic reconstruction, the proposed framework aims to recover a latent representation that is consistent with the target semantic distribution, thereby reducing the degradation caused by multi-user interference.

The proposed framework is trained in multiple stages. First, the semantic encoder and generative decoder are trained to establish a structured latent representation and reliable image reconstruction. Subsequently, the DDPM-based separator is trained using latent mixtures generated under different interference conditions. During separator training, the interference level is varied to improve robustness against different degrees of latent-space interference. Once training is completed, the semantic encoder and decoder are fixed, and the trained separator is used to recover the target latent representation from the received mixture. The following sections describe the major components of the proposed framework in detail, including latent representation and transmission, conditional DDPM-based latent separation, semantic reconstruction, and the training strategy of the proposed system.

3.2 Semantic Representation and Transmission

The proposed framework performs semantic transmission in a learned latent space. As defined in Chapter 2, the semantic encoder maps the input source $\mathbf{x}_k$ to a latent representation $\mathbf{z}_k$. In the proposed implementation, a VAE-based semantic encoder is employed to obtain a compact latent representation for subsequent transmission and latent separation [21].

3.2.1 Semantic Encoding

For each user $k \in \mathcal{U}$, the input image $\mathbf{x}_k$ is mapped into a low-dimensional latent space using a shared variational encoder. Given $\mathbf{x}_k$, the encoder produces the posterior parameters $\boldsymbol{\mu}_k$ and $\boldsymbol{\sigma}_k$, defining the approximate posterior distribution as [22]

$$q_\phi(\mathbf{z}|\mathbf{x}_k) = \mathcal{N}\left(\boldsymbol{\mu}_k, \mathrm{diag}(\boldsymbol{\sigma}_k^2)\right), \tag{3.3}$$

where $\boldsymbol{\mu}_k$ and $\boldsymbol{\sigma}_k$ denote the posterior mean and standard deviation predicted by the encoder, respectively, and ϕ denotes the encoder parameters. During variational representation learning, a latent sample can be obtained through the reparameterization operation

$$\mathbf{z}_k^{\mathrm{s}} = \boldsymbol{\mu}_k + \boldsymbol{\sigma}_k \odot \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tag{3.4}$$

where $\mathbf{z}_k^{\mathrm{s}}$ denotes a stochastic latent sample. The learned posterior is regularized toward a standard Gaussian prior through the Kullback–Leibler (KL) divergence during representation learning [22, 23].

For semantic transmission, however, the proposed communication pipeline does not use the randomly sampled latent $\mathbf{z}_k^{\mathrm{s}}$. Instead, the posterior mean is adopted as the deterministic semantic representation:

$$\mathbf{z}_k \triangleq \boldsymbol{\mu}_k = E_\phi^{(\mu)}(\mathbf{x}_k), \tag{3.5}$$

where $E_\phi^{(\mu)}(\cdot)$ denotes the mean output of the variational encoder. Although both $\boldsymbol{\mu}_k$ and $\boldsymbol{\sigma}_k$ are learned during variational representation learning, only $\boldsymbol{\mu}_k$ is used for semantic transmission. This avoids introducing additional sampling randomness into the communication pipeline and provides a deterministic latent target for the subsequent latent separation process. The resulting representation $\mathbf{z}_k$ is then globally scaled before channel transmission, as described in the following subsection.

3.2.2 Global Latent Scaling and Transmission

After semantic encoding, the deterministic latent representation $\mathbf{z}_k = \boldsymbol{\mu}_k$ is prepared for transmission. To maintain a consistent latent scale while preserving the statistical structure learned by the semantic encoder and generative decoder, a fixed global scaling factor is applied to the encoder output. Specifically, the transmitted latent representation is defined as

$$\tilde{\mathbf{z}}_k = \gamma\, \mathbf{z}_k = \gamma\, \boldsymbol{\mu}_k, \tag{3.6}$$

where $\gamma > 0$ denotes a fixed global scaling factor determined from the latent representations of the training data. Unlike sample-wise normalization, the same scaling factor is applied to all latent representations. This operation preserves the relative structure of the learned latent space and provides a consistent scale for subsequent channel transmission and latent separation.

The scaled latent representations are then transmitted simultaneously through the shared wireless channel according to the multi-user transmission model defined in (2.4). The resulting received latent mixture $\mathbf{y}_k$ serves as the observation for the user-specific latent separation process described in the following section.

3.3 Conditional DDPM-Based Latent Separation

After multi-user transmission, the receiver observes the latent mixture $\mathbf{y}_k$ defined in (2.4). To recover a specified semantic component from the received observation, a conditional DDPM is employed as the latent separator. The received mixture $\mathbf{y}_k$ provides observation guidance, while the target condition c_k specifies the semantic component to be recovered. The conditional separation mechanism, forward diffusion process, reverse denoising process, and training objective are described in the following subsections.

3.3.1 Conditional Latent Separation

After multi-user transmission, the receiver obtains the latent mixture $\mathbf{y}_k$ defined in (2.4). The received observation contains the desired semantic latent representation together with interference from other users and channel noise. Therefore, the desired latent representation cannot be directly obtained from $\mathbf{y}_k$.

To recover a specific semantic component from the received mixture, a conditional DDPM-based latent separator is employed. The separator uses

the received mixture $\mathbf{y}_k$ as the observation of the current transmission and a target condition c_k to specify which semantic component should be recovered. Accordingly, the latent separation problem is modeled as the conditional distribution

$$p_\theta\left(\tilde{\mathbf{z}}_k \mid \mathbf{y}_k, c_k\right), \tag{3.7}$$

where $\tilde{\mathbf{z}}_k$ denotes the globally scaled latent representation of the target user and θ denotes the parameters of the conditional denoising network.

The denoising network is expressed as

$$\hat{\boldsymbol{\epsilon}}_k = \epsilon_\theta\left(\tilde{\mathbf{z}}_{k,t}, t, \mathbf{y}_k, c_k\right), \tag{3.8}$$

where $\tilde{\mathbf{z}}_{k,t}$ denotes the noisy latent state at diffusion timestep t. The received mixture $\mathbf{y}_k$ provides information about the current transmission, while c_k identifies the target semantic component to be recovered.

Therefore, for the same received mixture, different target conditions can guide the reverse denoising process toward different semantic components. For example, given a received mixture containing two semantic sources, selecting c_1 or c_2 enables the separator to recover the latent representation corresponding to the desired source.

3.3.2 Conditional Reverse Denoising Process

The reverse diffusion process aims to progressively recover the target latent representation from a noisy diffusion state. Unlike the forward process, the reverse process explicitly utilizes the received latent mixture $\mathbf{y}_k$ and the target condition c_k to guide the denoising process. At each reverse diffusion step, the conditional denoising network in (3.8) predicts the noise component from the current noisy latent state. Based on the predicted noise, an estimate of the clean target latent representation is obtained as

$$\hat{\tilde{\mathbf{z}}}_{k,0} = \frac{1}{\sqrt{\bar{a}_t}}\left(\tilde{\mathbf{z}}_{k,t} - \sqrt{1-\bar{a}_t}\,\hat{\boldsymbol{\epsilon}}_k\right), \tag{3.9}$$

where $\bar{a}_t$ denotes the cumulative product of the diffusion coefficients defined in the forward diffusion process.

During inference, the clean target latent representation is unavailable. Therefore, the reverse process is initialized from Gaussian noise,

$$\tilde{\mathbf{z}}_{k,T} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tag{3.10}$$

and iteratively denoises the latent state from $t = T$ to $t = 1$. Throughout the reverse process, the received mixture $\mathbf{y}_k$ remains fixed and provides observation guidance, while the target condition c_k determines which semantic

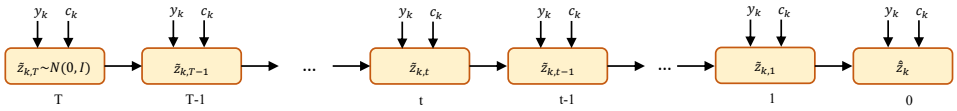


Figure 3.2: Conditional reverse denoising process for target-specific latent recovery. The received latent mixture $\mathbf{y}_k$ and the target condition c_k remain fixed and guide each reverse denoising step from $t = T$ to $t = 1$.

component is recovered. The overall conditional reverse denoising process is illustrated in Fig. 3.2.

After completing the reverse denoising process, the final latent state is taken as the recovered target latent representation,

$$\hat{\tilde{\mathbf{z}}}_k = \hat{\tilde{\mathbf{z}}}_{k,0}. \tag{3.11}$$

3.3.3 Training Objective

The conditional latent separator is trained to predict the Gaussian noise introduced during the forward diffusion process. For each training sample, a diffusion timestep t is randomly selected, and a noisy target latent $\tilde{\mathbf{z}}_{k,t}$ is generated from the clean target latent $\tilde{\mathbf{z}}_{k,0}$ according to (3.8). The denoising network predicts the added Gaussian noise conditioned on the received mixture $\mathbf{y}_k$ and the target condition c_k. The corresponding noise-prediction loss, hereafter referred to as the ϵ-loss, is defined for target k as

$$\mathcal{L}_{\epsilon}^{(k)} = \mathbb{E}_{t,\boldsymbol{\epsilon}_k}\left[\left\|\boldsymbol{\epsilon}_k - \epsilon_\theta\left(\tilde{\mathbf{z}}_{k,t}, t, \mathbf{y}_k, c_k\right)\right\|_2^2\right]. \tag{3.12}$$

This objective directly measures the discrepancy between the actual Gaussian noise $\boldsymbol{\epsilon}_k$ introduced during the forward diffusion process and the noise predicted by the conditional separator.

In addition to the ϵ-loss, a clean-latent reconstruction loss is introduced to directly constrain the recovery of the clean target latent representation. Using the estimated clean latent in (3.9), the reconstruction loss is defined as

$$\mathcal{L}_{z}^{(k)} = \mathbb{E}\left[\left\|\tilde{\mathbf{z}}_{k,0} - \hat{\tilde{\mathbf{z}}}_{k,0}\right\|_2^2\right]. \tag{3.13}$$

The DDPM training objective for target k is therefore formulated as

$$\mathcal{L}_{\text{DDPM}}^{(k)} = \mathcal{L}_{\epsilon}^{(k)} + \lambda_z \mathcal{L}_{z}^{(k)}, \tag{3.14}$$

where λ_z controls the contribution of the clean-latent reconstruction loss.

For multiple semantic domains, the objective is evaluated for each target condition and averaged to train a unified conditional latent separator. For the two-domain scenario considered in this work, the overall separation loss is expressed as

$$\mathcal{L}_{\text{sep}} = \frac{1}{2}\left(\mathcal{L}_{\text{DDPM}}^{(1)} + \mathcal{L}_{\text{DDPM}}^{(2)}\right). \tag{3.15}$$

3.4 Conditional WGAN-GP-Based Semantic Reconstruction

After the conditional DDPM-based latent separator recovers the desired semantic latent representation, the separated latent is mapped back to the image domain for semantic reconstruction. In the proposed framework, a conditional Wasserstein generative adversarial network with gradient penalty (WGAN-GP) is employed as the semantic reconstruction model [24, 25]. The generator is conditioned on the target semantic domain, allowing a shared generative model to reconstruct images from different source domains [26].

Let $\hat{\tilde{\mathbf{z}}}_k$ denote the separated latent representation obtained from the conditional reverse denoising process. Since a global scaling factor γ is applied to the semantic latent representation before transmission, the separated latent is first restored to its original scale as

$$\hat{\mathbf{z}}_k = \frac{\hat{\tilde{\mathbf{z}}}_k}{\gamma}, \tag{3.16}$$

where $\hat{\mathbf{z}}_k$ denotes the recovered semantic latent representation of target user k. The recovered latent representation is then provided to the conditional generator together with the target condition c_k:

$$\hat{\mathbf{x}}_k = G_\psi\left(\hat{\mathbf{z}}_k, c_k\right), \tag{3.17}$$

where $G_\psi(\cdot)$ denotes the conditional generator parameterized by ψ, c_k denotes the semantic-domain condition associated with user k, and $\hat{\mathbf{x}}_k$ denotes the reconstructed semantic source. The target condition c_k enables the same generator to reconstruct semantic information belonging to different source domains. In this work, the source domains correspond to different satellite image datasets, and the associated domain condition is provided to the generator together with the recovered latent representation.

To establish a latent space suitable for semantic reconstruction, the generator is initially pretrained using the conditional WGAN-GP objective with

randomly sampled Gaussian latent vectors. It is subsequently jointly fine-tuned with the VAE-based semantic encoder such that the deterministic posterior mean produced by the encoder becomes compatible with the generator latent space. The detailed training procedure is presented in Section 3.5.

3.5 Training Strategy

The proposed framework consists of multiple learning components with different optimization objectives. Jointly optimizing all components from random initialization may lead to unstable training, since the latent separator would continuously receive latent representations whose distribution changes as the semantic encoder and generative reconstruction model are updated. Therefore, a stage-wise training strategy is adopted to decouple semantic representation learning from latent interference separation. The overall training procedure consists of two main stages. In Stage I, the VAE-based semantic encoder and the conditional WGAN-GP-based reconstruction model are trained to establish a stable semantic latent space and reliable image reconstruction. In Stage II, the parameters of the pretrained semantic encoder and conditional generator are fixed, and only the parameters of the conditional DDPM-based latent separator are optimized. The overall two-stage training strategy is illustrated in Fig. 3.3.

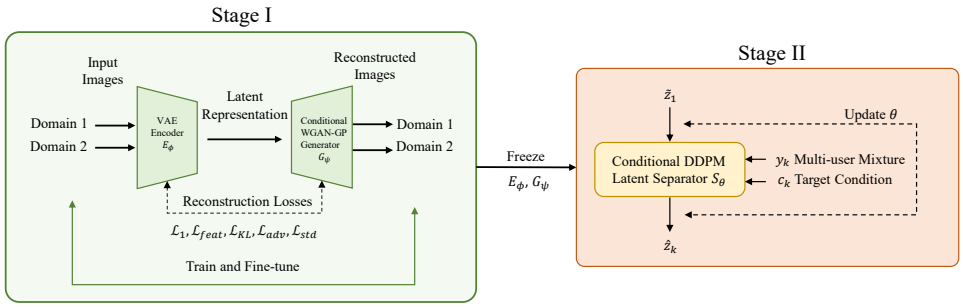


Figure 3.3: Two-stage training strategy of the proposed TASS framework. In Stage I, the semantic encoder and conditional generator are trained to establish a stable semantic latent space. In Stage II, the encoder and generator are fixed, and only the conditional DDPM-based latent separator is optimized using multi-user latent mixtures.

The parameter optimization and gradient flow of the two training stages are described in the following subsections. The complete Stage II training procedure is summarized in Algorithm 1.

3.5.1 Stage I: Semantic Representation Learning

The first training stage aims to establish a stable semantic latent space that preserves the information required for image reconstruction. This stage consists of two successive training phases: conditional WGAN-GP pretraining followed by joint fine-tuning of the VAE-based semantic encoder and the pretrained conditional generator. The conditional WGAN-GP is first pretrained to establish a generative latent space for the considered semantic domains using the WGAN-GP objective [25]. A random latent vector $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and a semantic-domain condition c_k are provided to the generator as

$$\tilde{\mathbf{x}} = G_\psi(\mathbf{z}, c_k), \tag{3.18}$$

where ψ denotes the generator parameters.

During conditional WGAN-GP pretraining, the critic is optimized according to

$$\mathcal{L}_D = \mathbb{E}\left[D_\omega(\tilde{\mathbf{x}}, c_k)\right] - \mathbb{E}\left[D_\omega(\mathbf{x}_k, c_k)\right] + \lambda_{\mathrm{GP}}\mathcal{L}_{\mathrm{GP}}, \tag{3.19}$$

where ω denotes the critic parameters and λ_{GP} is the gradient-penalty coefficient. The gradient penalty is defined as

$$\mathcal{L}_{\mathrm{GP}} = \mathbb{E}\left[\left(\|\nabla_{\bar{\mathbf{x}}} D_\omega(\bar{\mathbf{x}}, c_k)\|_2 - 1\right)^2\right], \tag{3.20}$$

where $\bar{\mathbf{x}}$ denotes an interpolated sample between real and generated samples. The generator is optimized according to

$$\mathcal{L}_G = -\mathbb{E}\left[D_\omega\left(G_\psi(\mathbf{z}, c_k), c_k\right)\right], \tag{3.21}$$

with the corresponding gradients

$$\mathbf{g}_\omega = \nabla_\omega \mathcal{L}_D, \qquad \mathbf{g}_\psi = \nabla_\psi \mathcal{L}_G. \tag{3.22}$$

The critic and generator parameters are updated using their corresponding optimizers.

After the conditional WGAN-GP has been pretrained, the VAE-based semantic encoder is introduced to map the input images into the pretrained generative latent space. For an input image $\mathbf{x}_k$, the encoder produces

$$\left(\boldsymbol{\mu}_k, \log \boldsymbol{\sigma}_k^2\right) = E_\phi(\mathbf{x}_k), \tag{3.23}$$

where ϕ denotes the encoder parameters. The posterior mean is adopted as the deterministic semantic latent representation used in the communication system,

$$\mathbf{z}_k \triangleq \boldsymbol{\mu}_k, \tag{3.24}$$

and the corresponding clean semantic reconstruction is obtained as

$$\hat{\mathbf{x}}_k^{\text{clean}} = G_\psi\left(\mathbf{z}_k, c_k\right). \tag{3.25}$$

During joint fine-tuning, the conditional critic is first updated while the encoder and generator are temporarily fixed. In addition to the reconstructed samples obtained from the encoder output, random-prior samples are generated to preserve the generative capability of the pretrained conditional generator. The critic objective is defined as

$$\begin{aligned}\mathcal{L}_D^{\text{joint}} = \frac{1}{2}\Big(&\mathbb{E}\left[D_\omega\left(\hat{\mathbf{x}}_k^{\text{clean}}, c_k\right)\right] + \mathbb{E}\left[D_\omega\left(\mathbf{x}_k^r, c_k\right)\right]\Big) \\ &- \mathbb{E}\left[D_\omega\left(\mathbf{x}_k, c_k\right)\right] + \lambda_{\text{GP}}\mathcal{L}_{\text{GP}},\end{aligned} \tag{3.26}$$

where

$$\mathbf{x}_k^r = G_\psi\left(\mathbf{z}^r, c_k\right), \qquad \mathbf{z}^r \sim \mathcal{N}\left(\mathbf{0}, \mathbf{I}\right). \tag{3.27}$$

The critic gradient is given by

$$\mathbf{g}_\omega = \nabla_\omega \mathcal{L}_D^{\text{joint}}, \tag{3.28}$$

and the critic parameters are updated while the encoder and generator remain fixed.

After the critic update, the critic parameters are fixed and the encoder and generator are jointly optimized. The pixel-domain reconstruction loss and feature-space reconstruction loss are respectively defined as

$$\mathcal{L}_{\text{L1}} = \mathbb{E}\left[\left\|\hat{\mathbf{x}}_k^{\text{clean}} - \mathbf{x}_k\right\|_1\right], \tag{3.29}$$

and

$$\mathcal{L}_{\text{feat}} = \mathbb{E}\left[\left\|F_\omega\left(\hat{\mathbf{x}}_k^{\text{clean}}\right) - F_\omega\left(\mathbf{x}_k\right)\right\|_1\right], \tag{3.30}$$

where $F_\omega(\cdot)$ denotes the feature representation extracted by the critic. The variational regularization term is defined as

$$\mathcal{L}_{\text{KL}} = -\frac{1}{2}\mathbb{E}\left[1 + \log \boldsymbol{\sigma}_k^2 - \boldsymbol{\mu}_k^2 - \boldsymbol{\sigma}_k^2\right], \tag{3.31}$$

while the adversarial objectives for reconstructed and random-prior samples are respectively given by

$$\mathcal{L}_{\text{adv,rec}} = -\mathbb{E}\left[D_\omega\left(\hat{\mathbf{x}}_k^{\text{clean}}, c_k\right)\right], \tag{3.32}$$

and

$$\mathcal{L}_{\text{adv,rand}} = -\mathbb{E}\left[D_\omega\left(\mathbf{x}_k^r, c_k\right)\right]. \tag{3.33}$$

To prevent the posterior standard deviation from becoming excessively small, an additional regularization term is introduced as

$$\mathcal{L}_{\text{std}} = \mathbb{E}\left[\max\left(0, \sigma_{\min} - \exp\left(\frac{1}{2}\log \boldsymbol{\sigma}_k^2\right)\right)\right], \tag{3.34}$$

where $\sigma_{\min}$ denotes the minimum desired posterior standard deviation. The overall encoder–generator objective used during joint fine-tuning is therefore

$$\begin{aligned}\mathcal{L}_{EG} = \lambda_{\text{L1}}\mathcal{L}_{\text{L1}} + \lambda_{\text{feat}}\mathcal{L}_{\text{feat}} + \beta_{\text{KL}}\mathcal{L}_{\text{KL}} \\ + \lambda_{\text{adv,rec}}\mathcal{L}_{\text{adv,rec}} + \lambda_{\text{adv,rand}}\mathcal{L}_{\text{adv,rand}} + \lambda_{\text{std}}\mathcal{L}_{\text{std}}.\end{aligned} \tag{3.35}$$

The corresponding gradients are

$$\mathbf{g}_\phi = \nabla_\phi \mathcal{L}_{EG}, \qquad \mathbf{g}_\psi = \nabla_\psi \mathcal{L}_{EG}. \tag{3.36}$$

The encoder and generator parameters are subsequently updated jointly using the Adam optimizer. After Stage I is completed, E_ϕ and G_ψ are fixed and no further parameter updates are applied to them during Stage II, thereby providing a stationary semantic latent space for training the conditional DDPM-based separator.

3.5.2 Interference-Level Randomization

To improve the robustness of the latent separator against varying interference conditions, the interference strength is randomized during Stage II training. Training with a single fixed interference coefficient may cause the separator to become specialized to a particular interference level and reduce its generalization ability under unseen interference conditions. Therefore, for each training iteration, the interference coefficient associated with the interfering user is independently sampled from a uniform distribution:

$$\alpha_i \sim \mathcal{U}\left(\alpha_{\min}, \alpha_{\max}\right), \qquad i \neq k, \tag{3.37}$$

where $\alpha_{\min}$ and $\alpha_{\max}$ denote the minimum and maximum interference amplitudes considered during training. The sampled α_i is then used in the multi-user channel model in (2.4) to construct the received latent mixture for the current training iteration.

By randomly varying α_i, the separator is exposed to latent mixtures with different interference strengths, ranging from weak to more challenging interference conditions. Consequently, the conditional separator is optimized over a range of interference levels rather than a single fixed value, encouraging the model to exploit the received observation $\mathbf{y}_k$ together with the target

condition c_k for target-specific latent recovery. This allows a single trained separator to operate under different interference strengths without retraining for each individual value of α_i. The complete training procedure of the proposed framework is summarized in Algorithm 1.

3.6 Inference Procedure

After training is completed, the semantic encoder, conditional generator, and conditional DDPM-based latent separator are fixed during inference. Given the received latent mixture $\mathbf{y}_k$ and the target condition c_k, the reverse diffusion process is performed to recover the desired user-specific latent representation. The recovered latent representation is then restored to its original scale and mapped to the image domain by the conditional generator. The complete inference procedure of the proposed TASS framework is summarized in Algorithm 2.

Algorithm 1 Two-Stage Training Procedure of the Proposed TASS Framework

Require: Datasets $\mathcal{D}_1, \mathcal{D}_2$; global scaling factor γ; diffusion steps T
Ensure: Trained semantic encoder E_ϕ, conditional generator G_ψ, and latent separator ϵ_θ

Stage I: Semantic Representation Learning
Pretrain the conditional WGAN-GP generator G_ψ and critic D_ω
Jointly fine-tune the semantic encoder E_ϕ and conditional generator G_ψ
Optimize the Stage-I training objectives until convergence
Freeze the trained semantic encoder E_ϕ and conditional generator G_ψ

Stage II: Latent Separator Training
while separator training has not converged **do**
    Sample $\mathbf{x}_1 \sim \mathcal{D}_1$ and $\mathbf{x}_2 \sim \mathcal{D}_2$
    Obtain posterior means $\boldsymbol{\mu}_1$ and $\boldsymbol{\mu}_2$ using the frozen encoder E_ϕ
    Set deterministic semantic latents $\mathbf{z}_1 \leftarrow \boldsymbol{\mu}_1$ and $\mathbf{z}_2 \leftarrow \boldsymbol{\mu}_2$
    Apply global latent scaling: $\tilde{\mathbf{z}}_1 \leftarrow \gamma \mathbf{z}_1$, $\tilde{\mathbf{z}}_2 \leftarrow \gamma \mathbf{z}_2$
    for $k \in \{1, 2\}$ **do**
        Set the interfering-user index $i \neq k$
        Sample interference coefficient $\alpha_i \sim \mathcal{U}(\alpha_{\min}, \alpha_{\max})$
        Construct the received latent mixture $\mathbf{y}_k$
        Set $\tilde{\mathbf{z}}_{k,0} \leftarrow \tilde{\mathbf{z}}_k$ and assign target condition c_k
        Sample diffusion timestep t and Gaussian noise $\boldsymbol{\epsilon}_{k,t}$
        Generate noisy target latent $\tilde{\mathbf{z}}_{k,t}$
        Predict diffusion noise $\hat{\boldsymbol{\epsilon}}_{k,t}$ conditioned on $(\tilde{\mathbf{z}}_{k,t}, t, \mathbf{y}_k, c_k)$
        Estimate clean target latent $\hat{\tilde{\mathbf{z}}}_{k,0}$
        Compute target-specific loss $\mathcal{L}_{\text{DDPM}}^{(k)}$
    end for
    Compute total separation loss $\mathcal{L}_{\text{sep}}$
    Compute separator gradient $\mathbf{g}_\theta \leftarrow \nabla_\theta \mathcal{L}_{\text{sep}}$
    Update only the separator parameters θ using Adam
end while
return E_ϕ, G_ψ, and ϵ_θ

Algorithm 2 Inference Procedure of the Proposed TASS Framework

Require: Received latent mixture $\mathbf{y}_k$; target condition c_k; trained conditional separator ϵ_θ; pretrained conditional generator G_ψ; global scaling factor γ; diffusion steps T

Ensure: Reconstructed target semantic source $\hat{\mathbf{x}}_k$

Sample the initial latent state $\tilde{\mathbf{z}}_{k,T} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

for $t = T, T-1, \ldots, 1$ **do**

    Predict the diffusion noise

$$\hat{\boldsymbol{\epsilon}}_k \leftarrow \epsilon_\theta \left(\tilde{\mathbf{z}}_{k,t}, t, \mathbf{y}_k, c_k \right)$$

    Compute the reverse-process mean $\boldsymbol{\mu}_\theta$

    Sample the next latent state $\tilde{\mathbf{z}}_{k,t-1}$

end for

Set the recovered scaled latent representation $\hat{\tilde{\mathbf{z}}}_k \leftarrow \tilde{\mathbf{z}}_{k,0}$

Restore the original latent scale $\hat{\mathbf{z}}_k \leftarrow \hat{\tilde{\mathbf{z}}}_k / \gamma$

Reconstruct the target semantic source $\hat{\mathbf{x}}_k \leftarrow G_\psi \left(\hat{\mathbf{z}}_k, c_k \right)$

return $\hat{\mathbf{x}}_k$

Chapter 4

Performance Evaluation

4.1 Experimental Setup

4.1.1 Datasets and Simulation Settings

The performance of the proposed semantic communication framework is evaluated using the EuroSAT and DeepGlobe datasets. Both datasets contain remote-sensing images but exhibit different visual and semantic characteristics, which enables the proposed method to be evaluated across different source distributions. All input images are resized to 64×64 pixels before being processed by the semantic encoder.

For each user, the input image is encoded into a 256-dimensional latent representation. To simulate multi-user transmission, the latent representation of the target user is superimposed with those of the interfering users. To simulate multi-user transmission, the latent representations of the target and interfering users are superimposed according to the multi-user interference model defined in Chapter 2. Unless otherwise specified, the same model configuration is adopted throughout the experiments.

Unless otherwise specified, the same model configuration is adopted throughout the experiments. The DDPM-based separator uses 1000 diffusion steps with a linear noise schedule ranging from 1×10^{-4} to 2×10^{-2}. The model is trained using the Adam optimizer with a learning rate of 2×10^{-4} and a batch size of 32.

Several experimental conditions are considered to evaluate different aspects of the proposed framework. First, the SNR is varied from 0 to 30 dB to investigate robustness against channel noise. Second, the interference amplitude α is varied to evaluate the performance under different interference strengths. The number of simultaneous interfering users is further varied from one to five to investigate the scalability of the proposed separator.

Table 4.1: Simulation and model parameters.

Parameter	**Setting**
Dataset	EuroSAT / DeepGlobe
Image resolution	64×64
Latent dimension	256
Global latent scaling factor c	0.1701
Semantic Representation Learning	
GAN pretraining epochs	30
Joint VAE–GAN training epochs	150
Joint training batch size	16
Encoder learning rate	1×10^{-4}
Generator learning rate	5×10^{-5}
Critic learning rate	1×10^{-4}
Gradient penalty λ_{GP}	10
Maximum KL weight $\beta_{\mathrm{KL,max}}$	1×10^{-4}
Latent Separator Training	
DDPM training epochs	100
Number of diffusion steps T	400
DDPM learning rate	2×10^{-4}
DDPM batch size	32
Clean-latent loss weight λ_{x_0}	0.2
Minimum diffusion variance β_1	1×10^{-4}
Maximum diffusion variance β_T	2×10^{-2}
SNR range	0–30 dB
Interference amplitude α	0–2.0

Finally, AWGN, Rayleigh, and Rician channel models are considered for performance comparison under different channel conditions.

The main simulation parameters are summarized in Table 4.1.

4.1.2 Evaluation Metric

The reconstruction quality is evaluated using the peak signal-to-noise ratio (PSNR), which measures the pixel-wise fidelity between the original image $\mathbf{x}$ and the reconstructed image $\hat{\mathbf{x}}$. PSNR is defined as

$$\mathrm{PSNR} = 10 \log_{10} \left(\frac{MAX^2}{\mathrm{MSE}} \right), \tag{4.1}$$

where MAX denotes the maximum possible pixel value and MSE represents the mean squared error between the original and reconstructed images. A higher PSNR value indicates better reconstruction fidelity.

4.2 Convergence Analysis

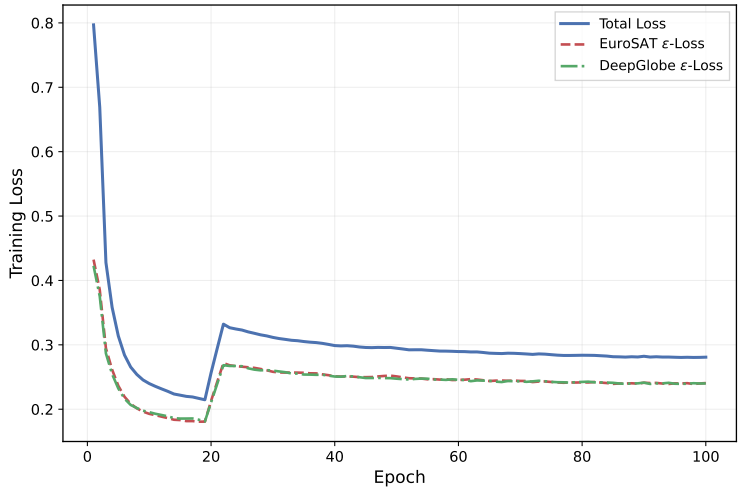
Before evaluating the reconstruction performance of the proposed semantic separator, its convergence behavior during training is first examined. Figure 4.1 presents the training losses of the conditional DDPM-based latent separator over 100 epochs. The overall separation loss $\mathcal{L}_{\text{sep}}$ and the noise-prediction (ϵ) losses $\mathcal{L}_{\epsilon}^{(k)}$ for EuroSAT and DeepGlobe are shown in Fig. 4.1(a), while the corresponding clean-latent (x_0) reconstruction losses $\mathcal{L}_{z}^{(k)}$ are shown in Fig. 4.1(b). These loss terms are defined in Section 3.3.3.

As shown in Fig. 4.1(a), the overall separation loss and the ϵ-losses decrease rapidly during the initial training stage. The overall separation loss decreases from approximately 0.8 to around 0.2 within the first 20 epochs. Similar convergence behavior is observed for the EuroSAT and DeepGlobe ϵ-losses, indicating that the separator progressively improves its ability to predict the Gaussian noise introduced during the forward diffusion process for both semantic domains.

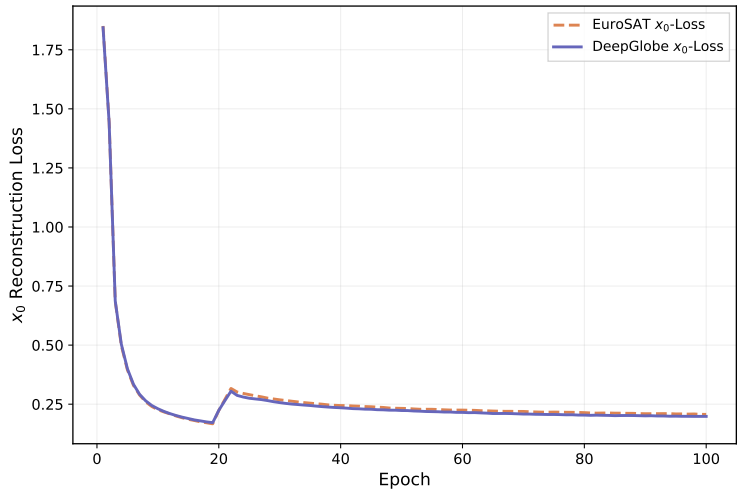
A temporary increase in the losses can be observed at approximately the 20th epoch. This behavior corresponds to the transition from the initial clean curriculum to the full interference-and-noise training condition. During the initial curriculum, the separator is trained under relatively mild interference without channel noise. After the transition, the training samples include stronger randomized interference and AWGN, resulting in a temporary increase in optimization difficulty. The losses subsequently decrease and gradually stabilize. In the later training stage, the overall separation loss $\mathcal{L}_{\text{sep}}$ converges to approximately 0.28, while the ϵ-losses of both semantic domains stabilize at approximately 0.24.

A similar convergence behavior is observed for the clean-latent reconstruction loss in Fig. 4.1(b). The $\mathcal{L}_{z}^{(k)}$ values decrease rapidly during the initial training stage and reach approximately 0.17 before the curriculum transition. Following the temporary increase at approximately the 20th epoch, the reconstruction losses decrease again and eventually stabilize at approximately 0.20. The similar convergence trends observed for EuroSAT and DeepGlobe indicate that the conditional separator learns to recover the target latent representations of both semantic domains without a significant training imbalance between them.

Overall, the convergence results show that the conditional DDPM-based latent separator reaches a stable training state after adapting to the full interference-and-noise training condition. The following sections evaluate the reconstruction performance of the converged model under different channel and multi-user interference conditions.



(a) Training loss



(b) x_0 reconstruction loss

Figure 4.1: Convergence behavior of the proposed DDPM-based semantic separator during training.

4.3 Preliminary Observations

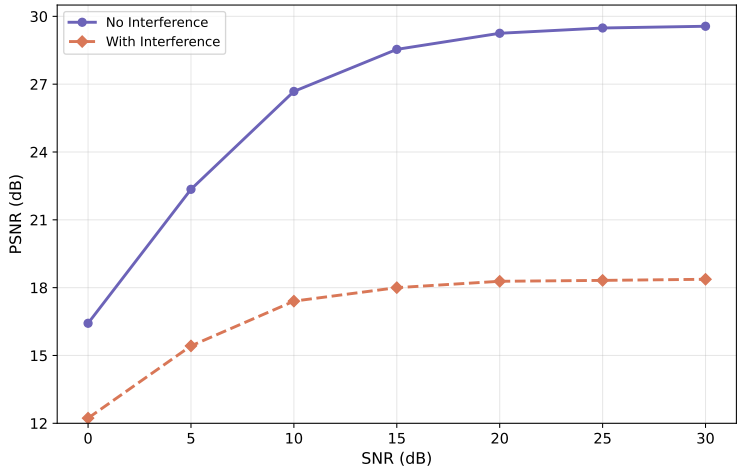
Before introducing the proposed interference mitigation method, we first examine the effect of inter-user semantic interference on semantic reconstruction. This preliminary experiment is conducted without applying any latent separation or interference cancellation technique.

Two transmission conditions are considered. In the no-interference case,

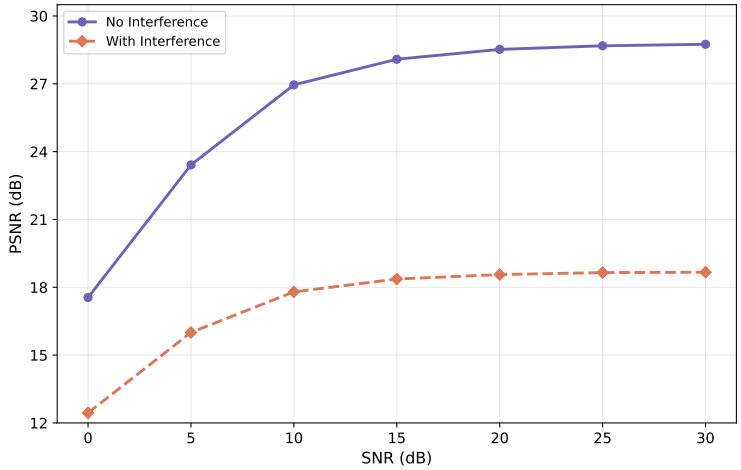
only the target user's latent representation is transmitted through the wireless channel. In the interfered case, the latent representation of an additional user is superimposed on the target latent representation according to the system model in (2.4). The received latent representation is directly reconstructed by the semantic decoder without any additional processing.

Figure 4.2 compares the reconstruction performance under the two transmission conditions over different SNRs. For both datasets, the reconstruction quality in the no-interference case improves substantially as the SNR increases. In contrast, the interfered case remains significantly degraded even at relatively high SNRs. This observation indicates that improving the channel SNR alone cannot eliminate the degradation caused by inter-user semantic interference.

These preliminary observations demonstrate the need for a mechanism that can identify and recover the desired user-specific latent representation from a received semantic mixture. This motivates the interference mitigation method developed in Chapter 3.



(a) EuroSAT



(b) DeepGlobe

Figure 4.2: Preliminary comparison of semantic reconstruction performance with and without inter-user interference.

4.4 Effectiveness of the Proposed Semantic Separator

After confirming the convergence of the proposed DDPM-based semantic separator, its effectiveness in recovering the target semantic source is evaluated. To isolate the contribution of the separator, three reconstruction conditions are considered: clean latent transmission, reconstruction without semantic

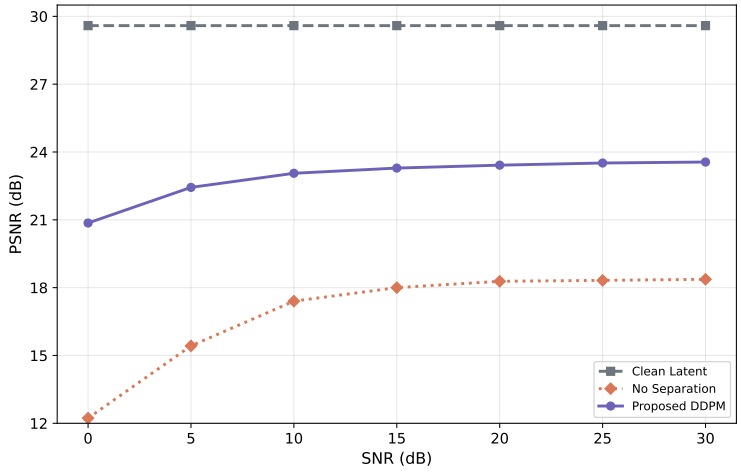
separation, and reconstruction using the proposed DDPM-based separator. The clean latent case serves as a reference for the reconstruction quality in the absence of multi-user interference, while the no-separation case directly reconstructs the received latent mixture without applying the proposed separator.

Figure 4.3 compares the PSNR performance under different SNR conditions for the EuroSAT and DeepGlobe datasets. For both datasets, the clean latent case maintains the highest PSNR over the entire SNR range, representing the reference reconstruction performance of the semantic encoder and generator. In contrast, the no-separation case exhibits a substantial degradation, particularly at low SNRs, due to the combined effects of channel noise and multi-user interference.

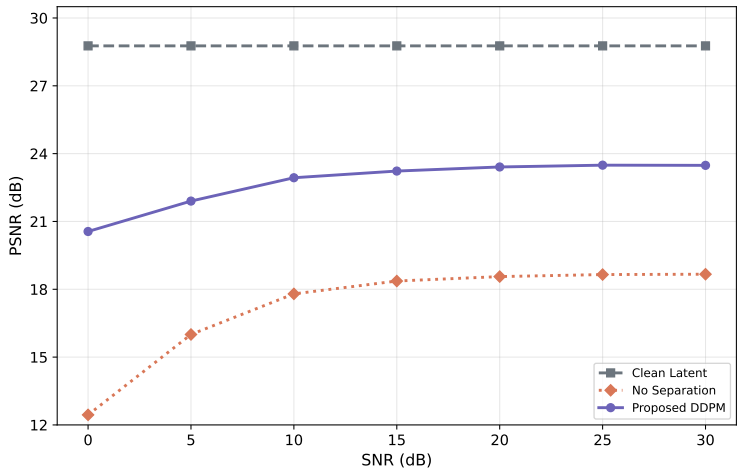
With the proposed DDPM-based semantic separator, a clear improvement over the no-separation case is observed for both datasets. As the SNR increases, the PSNR of the proposed method improves and gradually approaches a stable level. For EuroSAT, the proposed method increases from approximately 21 dB at 0 dB SNR to approximately 23.5 dB at high SNRs. A similar trend is observed for DeepGlobe, where the PSNR increases from approximately 20.5 dB to 23.5 dB. In comparison, the no-separation case remains considerably lower over most of the evaluated SNR range.

The performance gap between the proposed method and the clean latent reference indicates that semantic reconstruction is still affected by the information loss introduced by interference and the separation process. Nevertheless, the consistent improvement over the no-separation case demonstrates that the proposed DDPM-based separator can effectively suppress interfering semantic components and recover a latent representation that is more suitable for reconstructing the target semantic source.

These results verify the fundamental effectiveness of the proposed semantic separator. The following sections further investigate its robustness by increasing the interference strength and the number of simultaneous interferers.



(a) EuroSAT



(b) DeepGlobe

Figure 4.3: PSNR performance of the proposed DDPM-based separator under multi-user interference.

4.5 Impact of Interference Strength

After verifying the effectiveness of the proposed semantic separator, we further investigate its robustness against different levels of multi-user interference. The interference strength is controlled by the interference amplitude α, which determines the contribution of the interfering latent representation to the received latent mixture. A larger value of α therefore corresponds to stronger multi-user interference.

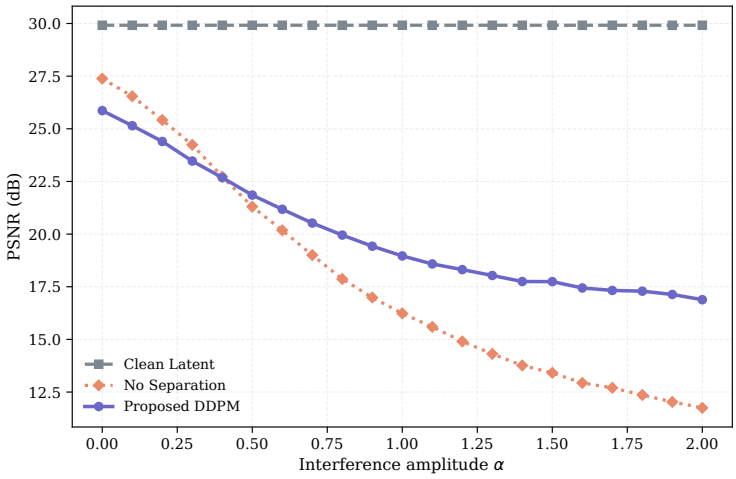
Figure 4.4 presents the PSNR performance of the proposed DDPM-based separator under different interference amplitudes for the EuroSAT and DeepGlobe datasets. The clean latent case is included as a reference, while the no-separation and proposed DDPM cases are compared to evaluate the effectiveness of semantic separation as the interference strength increases.

As shown in Fig. 4.4a, the reconstruction quality of EuroSAT gradually decreases as the interference amplitude increases. Under relatively weak interference, the no-separation case achieves comparable or slightly higher PSNR than the proposed method. This is because the received latent representation remains relatively close to the target latent when the interference component is weak, and applying an additional separation process may introduce reconstruction distortion. However, as the interference amplitude increases, the PSNR of the no-separation case decreases substantially, whereas the proposed DDPM-based separator exhibits a considerably slower degradation.

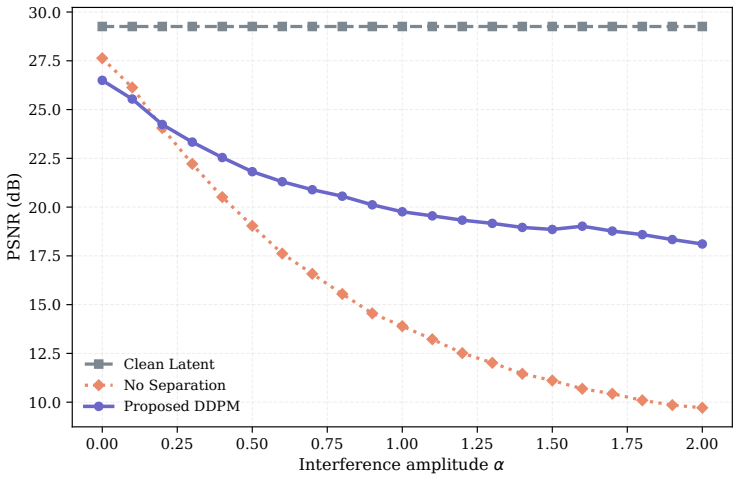
A similar trend can be observed for DeepGlobe in Fig. 4.4b. The performance difference between the two methods becomes increasingly evident as α increases. In particular, under strong interference, the proposed separator maintains substantially higher reconstruction quality than direct reconstruction without separation. This demonstrates that the proposed method is capable of recovering useful target semantic information even when the interfering latent component becomes dominant in the received mixture.

The results also reveal an important trade-off in the use of the semantic separator. When interference is sufficiently weak, direct reconstruction may already preserve most of the target information, and semantic separation provides limited benefit. In contrast, as the interference strength increases, the advantage of the proposed separator becomes increasingly significant. These observations demonstrate that the proposed DDPM-based separator is particularly effective in moderate-to-strong interference conditions.

Having evaluated the effect of interference strength, the next section further investigates the scalability of the proposed method when multiple interfering users are simultaneously present.



(a) EuroSAT



(b) DeepGlobe

Figure 4.4: PSNR performance of the proposed DDPM-based separator under different interference levels.

4.6 Impact of the Number of Interferers

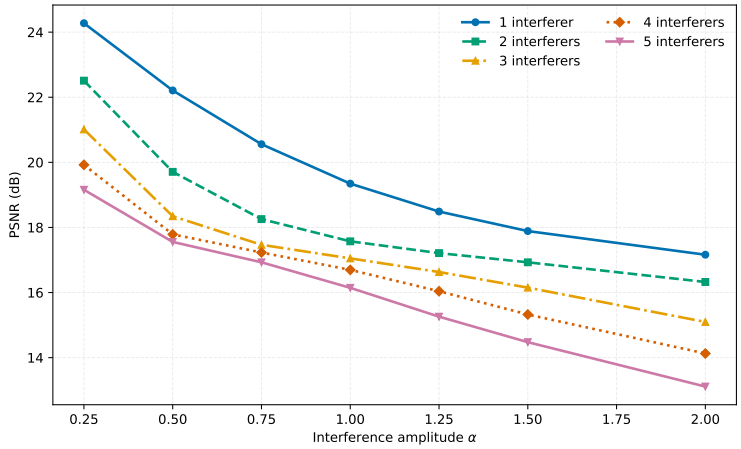
In addition to the interference strength, the number of simultaneous interfering users is another important factor affecting the reconstruction performance in a multi-user semantic communication system. Therefore, the scalability of the proposed DDPM-based semantic separator is further evaluated by increasing the number of interfering users from one to five.

Figure 4.5 presents the PSNR performance under different numbers of interferers and interference amplitudes for the EuroSAT and DeepGlobe datasets. For all evaluated cases, the reconstruction quality decreases as the interference amplitude α increases. Furthermore, increasing the number of interfering users results in an additional reduction in PSNR, since a larger amount of interfering semantic information is superimposed on the target latent representation.

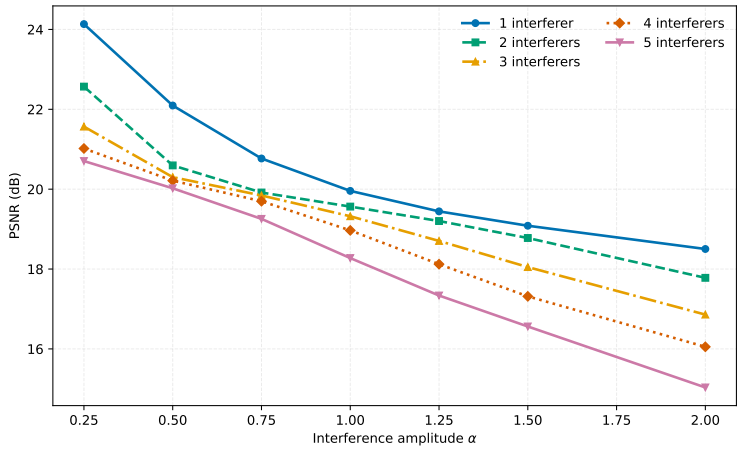
For EuroSAT, as shown in Fig. 4.5a, the case with a single interferer consistently achieves the highest PSNR over the evaluated range of α. As additional interferers are introduced, the PSNR gradually decreases. The degradation becomes more pronounced at larger values of α, demonstrating the combined effect of interference strength and the number of interfering users on the reconstruction quality.

A similar behavior is observed for DeepGlobe in Fig. 4.5b. The reconstruction performance decreases progressively as the number of interferers increases from one to five. Nevertheless, the proposed separator continues to recover the target semantic source under all evaluated multi-user configurations, although with reduced reconstruction fidelity as the aggregate interference becomes stronger.

These results demonstrate that both the interference strength and the number of simultaneous interferers have a significant impact on the reconstruction performance. While the proposed DDPM-based separator remains applicable when multiple interfering users are present, its performance gradually degrades as the multi-user interference becomes more severe. This observation highlights the trade-off between reconstruction quality and the level of multi-user interference in the proposed semantic communication framework.



(a) EuroSAT



(b) DeepGlobe

Figure 4.5: PSNR performance of the proposed DDPM-based separator under different numbers of interferers.

4.7 Performance under Different Channel Models

After evaluating the proposed semantic separator under different multi-user interference conditions, its performance is further compared with the VAE-based baseline under different wireless channel models. Three representative channel conditions are considered, including additive white Gaussian noise (AWGN), Rayleigh fading, and Rician fading. The comparison is conducted over an SNR range from 0 to 30 dB.

Figure 4.6 presents the average PSNR performance of the VAE-based baseline and the proposed method under the three channel models. In general, the reconstruction quality of both methods improves as the SNR increases. However, the proposed method consistently achieves higher PSNR than the VAE-based baseline over the evaluated SNR range and channel conditions.

Under the AWGN channel, the proposed method achieves the highest reconstruction quality among the evaluated cases. Its PSNR increases rapidly at low-to-medium SNRs and gradually saturates as the SNR increases. A similar trend is observed under the Rayleigh and Rician fading channels, although the achievable reconstruction quality is lower than that under AWGN due to the additional distortion introduced by channel fading.

The VAE-based baseline exhibits a similar dependence on the channel condition, but its PSNR remains lower than that of the proposed method. The performance difference is maintained across the evaluated SNR range, indicating that the improvement achieved by the proposed method is not limited to a specific channel model. In particular, the proposed method maintains its reconstruction advantage under both Rayleigh and Rician fading conditions, demonstrating its applicability beyond the AWGN channel.

It can also be observed that the performance improvement becomes smaller at relatively high SNRs. This behavior indicates that, as the channel noise becomes less dominant, the reconstruction quality is increasingly limited by the semantic representation and reconstruction process rather than by the channel noise itself. Nevertheless, the proposed method consistently provides improved reconstruction fidelity compared with the VAE-based baseline under all considered channel conditions.

Overall, the results demonstrate that the proposed semantic communication framework provides improved reconstruction performance over the VAE-based baseline and maintains this advantage under different wireless channel models.

4.8 Preliminary Hardware Prototype

In addition to the simulation-based evaluation, a preliminary hardware prototype was developed to examine the feasibility of multi-user semantic transmission in a practical wireless environment. Figure 4.7 shows the experimental setup, which consists of two transmitters and one receiver equipped with omnidirectional antennas. The two transmitters were placed approximately 0.6 m apart, while the transmitter-to-receiver distance was approximately 0.5 m.

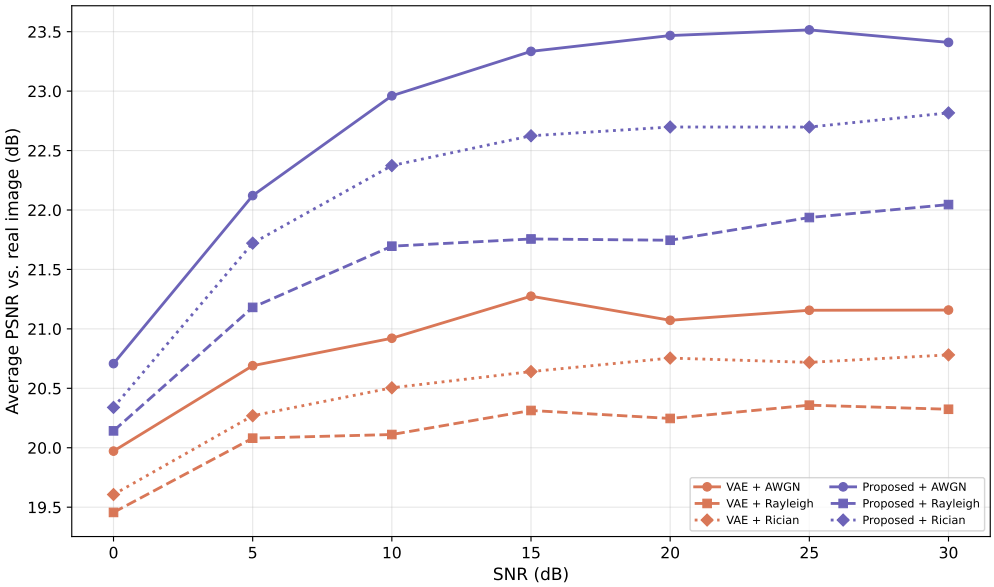
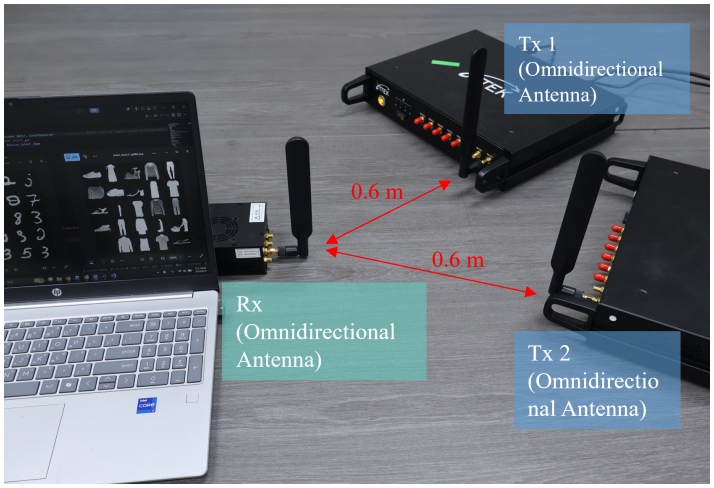


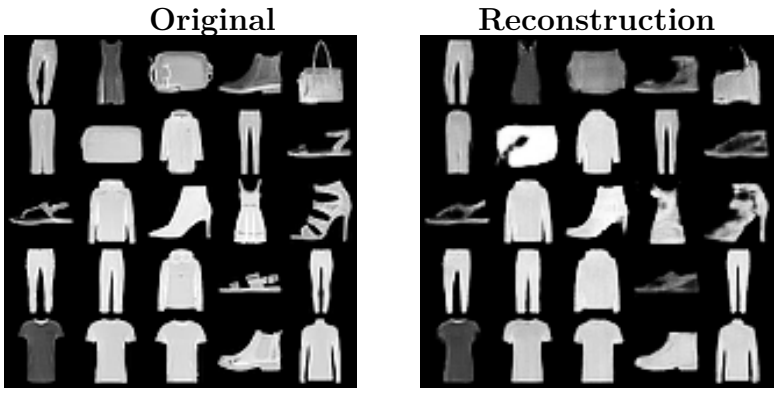
Figure 4.6: PSNR performance of the VAE-based baseline and the proposed method under different channel conditions.

As a preliminary demonstration, FashionMNIST and MNIST images were transmitted through the prototype system and reconstructed at the receiver. Example original and reconstructed images are shown in Fig. 4.7(b) and Fig. 4.7(c), respectively. The average PSNR values of the reconstructed FashionMNIST and MNIST images were 19.17 dB and 19.13 dB, respectively. These preliminary results demonstrate the feasibility of transmitting and reconstructing semantic information using the developed hardware platform.

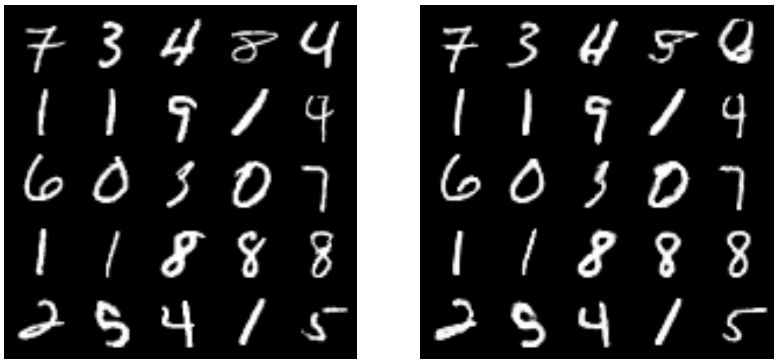
Due to the limited development time, the complete proposed DDPM-based semantic separation framework was not fully integrated into the hardware prototype. Therefore, the prototype results are presented as a preliminary proof-of-concept rather than a complete experimental validation of the proposed method. Full integration and evaluation of the proposed semantic separator in practical wireless environments are left for future work.



(a) Two-transmitter prototype setup.



(b) FashionMNIST images (average PSNR: 19.17 dB).



(c) MNIST images (average PSNR: 19.13 dB).

Figure 4.7: Preliminary hardware prototype of the two-transmitter semantic communication system: (a) experimental setup, (b) FashionMNIST reconstruction, and (c) MNIST reconstruction.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

This thesis investigated a multi-user semantic communication framework for image transmission in the presence of inter-user semantic interference. Unlike conventional communication systems that primarily focus on recovering transmitted symbols or bits, the considered framework aims to recover the semantic information associated with a desired source. In a multi-user environment, latent representations from multiple users may be superimposed during transmission, making user-specific semantic recovery challenging.

To address this problem, a Target-Aware Semantic Separation (TASS) framework based on a conditional DDPM was developed to recover the desired latent representation from the received latent mixture. The proposed framework integrates semantic encoding, wireless transmission, latent separation, and generative reconstruction into a unified semantic communication architecture. By conditioning the separator on both the received latent mixture and the target condition, TASS progressively recovers the latent representation corresponding to the desired semantic source through the reverse diffusion process.

The proposed framework was evaluated using the EuroSAT and DeepGlobe remote-sensing image datasets under different communication and interference conditions. The evaluation considered different signal-to-noise ratios, interference strengths, numbers of interfering users, and wireless channel models. The experimental results showed that the proposed separator consistently improves semantic reconstruction performance compared with the no-separation case and exhibits greater robustness as inter-user interference becomes stronger. The results also demonstrated that the proposed method remains effective when the number of interferers increases and under different

channel conditions, including AWGN, Rayleigh fading, and Rician fading.

Comparisons with the VAE-based baseline further showed that the proposed framework maintains improved reconstruction performance under different wireless channel conditions. These results indicate that diffusion-based latent separation provides a promising approach for recovering user-specific semantic information from mixed latent representations, particularly when inter-user interference becomes a significant factor in the received signal.

In addition to the simulation-based evaluation, a preliminary hardware prototype was established to investigate the feasibility of semantic transmission in a practical multi-transmitter wireless environment. The prototype demonstrated end-to-end semantic transmission and reconstruction using physical wireless devices. However, the proposed DDPM-based separator has not yet been fully integrated into the hardware platform. Therefore, the current prototype serves primarily as a proof-of-concept and provides a foundation for future over-the-air validation of the complete TASS framework.

Overall, this work demonstrates the feasibility of performing user-specific semantic separation directly in the latent domain for multi-user semantic communication. By combining target-conditioned diffusion-based separation with generative semantic reconstruction, the proposed framework provides an alternative approach to mitigating inter-user semantic interference and establishes a foundation for more robust multi-user generative semantic communication systems.

5.2 Future Work

Several directions can be considered for extending the proposed framework.

First, the preliminary hardware prototype can be further developed into a complete over-the-air implementation of the proposed TASS framework. Future experiments may integrate the semantic encoder, wireless transmission modules, conditional DDPM separator, and generative decoder into an end-to-end hardware platform. This would enable evaluation under practical impairments such as synchronization errors, channel estimation errors, hardware imperfections, and time-varying wireless channels.

Second, the proposed framework can be extended to larger and more dynamic multi-user scenarios. Although the experiments in this thesis investigate the effect of multiple interfering semantic sources, practical wireless networks may contain an unknown and time-varying number of users. Developing a separator that can adapt to varying numbers of interferers and more complex interference patterns would therefore be an important direction for future research.

Third, the computational complexity and inference latency of the DDPM-based separator should be further investigated. The iterative reverse diffusion process requires multiple denoising steps, which introduces additional computational cost compared with conventional feed-forward reconstruction methods. Future work may investigate accelerated diffusion sampling, reduced-step diffusion processes, consistency-based generative models, or other efficient architectures to enable low-latency and potentially real-time semantic separation.

Finally, future research may extend the proposed framework beyond remote-sensing image transmission. Other image domains, as well as semantic modalities such as text, audio, and multimodal information, may be considered to evaluate the generalization capability of target-aware semantic separation. Such extensions would provide a broader understanding of diffusion-based semantic separation and its potential applications in future multi-user semantic communication networks.

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